NOBEL LAUREATES BY THE NUMBER

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INTRODUCTION

Background information: Established by Swedish inventor and businessman Alfred Bernhard Nobel in 1895, the Nobel prize has six categories and as of 2019, has been awarded a total of 923 times to individual laureates.

This project aims to trace patterns behind the laureates throughout the years by exploring and visualizing factors such as award category, prize amount, award year, age, gender, birth country, and academic affiliation.

QUESTION

- What are the average demographics of a Nobel laureate?
- What is the gender gap like across all categories?
- What are some noticeable trends with the prize amount? Does it increase over time?

EXPERIMENTAL APPROACH

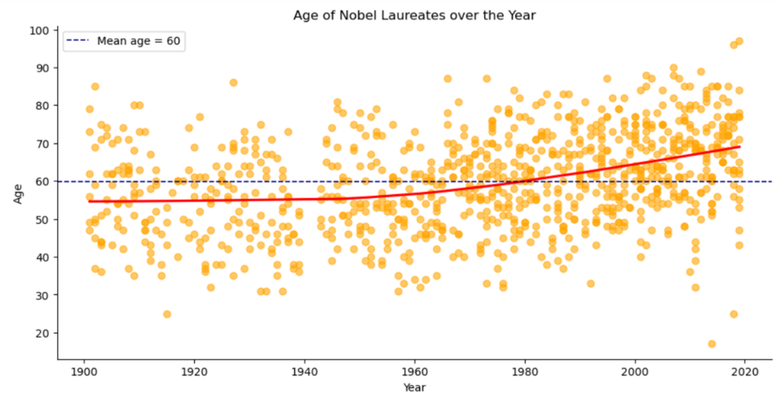
Data Source: Nobel Prize Dataset (1901-2019) from Kaggle (950 rows x 52 columns)

Steps:

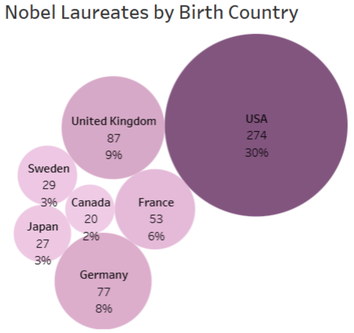
- Data wrangling: Eliminate unnecessary columns, handle invalid data.
- Data visualization
- Statistical summaries and hypothesis testing

Tools: R, Python, Tableau.

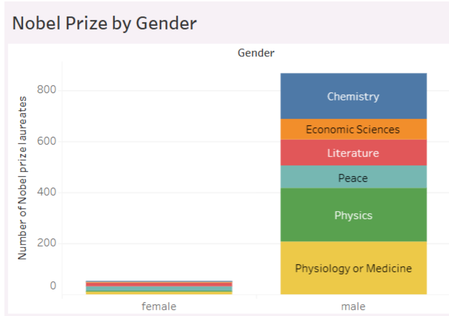
RESULTS



Nobel Laureates by Birth Country

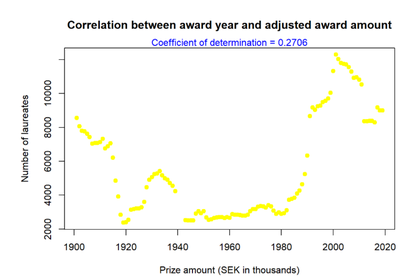


Nobel Prize by Gender



Male: 869 Female: 54

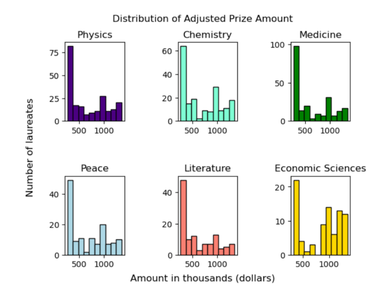
Top 10 Institutions Ranked by Nobel prize winners



Statistical analysis shows that the p-value < 2.2e-16 and the R-squared is 0.2706.

DISCUSSION

- Use logically weighted scatterplot smoothing (LOWESS)
- Regression analysis shows that there's a statistically significant positive but not linear relationship between the year and the inflation-adjusted prize amount (p-value < 0.05).



	Coefficient	p-value
Physics	0.4084	1.2137e-18
Chemistry	0.3997	1.1663e-15
Physiology	0.4078	4.4173e-19
Peace	0.3272	2.3426e-08
Literature	0.2398	0.0001373
Economic Sciences	0.4281	1.2088e-08

- Kendall's Tau correlation is considered to be moderate if the correlation value is between 0.29 and 0.49.
- Year and prize amount have the strongest relationship in economic sciences.

CONCLUSION

- The average Nobel laureate is male, US-born, and 60 years old.
- Women make up approximately 5.9% of all laureates, with Peace has the highest number of female recipients (17), followed by Literature (15).
- While year is a valid indicator of the adjusted prize amount, their relationship is non-linear. The Kendall's Tau correlation demonstrates that the strength of this relationship varies among six categories.